Week 1, Day 1

How Networks Work: Foundations for Cybersecurity

Goal of today: understand how data travels from your laptop to a server, so that when you later attack or defend something, you know exactly which layer you are touching.

Note: every abbreviation is written with its full form. A complete glossary is on the last page.

1. Why security people start with networking

Almost every attack travels over a network. Port scanning, sniffing, phishing links, web exploits and malware callbacks all rely on the concepts below. If you do not understand them, tools like Nmap (Network Mapper) and Wireshark are just buttons.

2. The layered model

Networking is split into layers. Each layer does one job and trusts the layer below it. Two models are used: OSI (Open Systems Interconnection, 7 layers, for teaching and vocabulary) and TCP/IP (Transmission Control Protocol / Internet Protocol, 4 layers, how the internet really works).

OSI layer	TCP/IP layer	What it does	Examples	Attack example
7 Application	Application	Software-level protocols	HTTP (HyperText Transfer Protocol), DNS (Domain Name System), FTP (File Transfer Protocol), SSH (Secure Shell), SMTP (Simple Mail Transfer Protocol)	SQL (Structured Query Language) injection, XSS (Cross-Site Scripting), phishing
6 Presentation	Application	Encoding, encryption	TLS (Transport Layer Security), JPEG (Joint Photographic Experts Group), JSON (JavaScript Object Notation)	SSL (Secure Sockets Layer) stripping, weak ciphers
5 Session	Application	Manages sessions	Cookies, tokens	Session hijacking
4 Transport	Transport	End-to-end delivery, ports	TCP (Transmission Control Protocol), UDP (User Datagram Protocol)	SYN (Synchronize) flood, port scanning
3 Network	Internet	Addressing and routing	IP (Internet Protocol), ICMP (Internet Control Message Protocol)	IP spoofing, ping sweeps
2 Data Link	Network access	Delivery on local network	Ethernet, Wi-Fi (Wireless Fidelity), MAC (Media Access Control) address, ARP (Address Resolution Protocol)	ARP spoofing, MAC flooding
1 Physical	Network access	Bits on the wire	Cables, radio	Cable tapping, jamming

Memory trick (top to bottom): All People Seem To Need Data Processing.

3. Encapsulation: how data is wrapped

When you open a website, your data is wrapped layer by layer on the way out, and unwrapped on the way in:

```
Application data  : GET /index.html
+ Transport header: source port 51514, destination port 443   -> segment
+ IP header       : source IP 192.168.1.10, dest IP 93.184.x.x -> packet
+ Ethernet header : source MAC, destination MAC                -> frame
Then the frame is sent as bits.
(IP = Internet Protocol, MAC = Media Access Control)
```

4. IP (Internet Protocol) addresses

- **IPv4 (Internet Protocol version 4):** four numbers from 0 to 255, like 192.168.1.10 (32 bits).
- **Public IP:** reachable on the internet. **Private IP:** used inside homes and offices. The private ranges are 10.0.0.0/8, 172.16.0.0/12 and 192.168.0.0/16.
- **NAT (Network Address Translation):** your router translates many private IPs into one public IP.
- **Localhost:** 127.0.0.1 always points to your own machine.
- **MAC (Media Access Control) address:** the hardware address of a network card. It is used only inside the local network, while the IP (Internet Protocol) address is used across networks.
- **Default gateway:** the router your device sends traffic to when the destination is outside your network.

5. Ports and common services

An IP (Internet Protocol) address finds the machine. A **port** (0 to 65535) finds the program on that machine. An **open port** means a service is listening, and each open service is a possible attack surface. This is why Nmap (Network Mapper) matters later.

Port	Protocol	Service (full form)	Why it matters in security
21	TCP	FTP (File Transfer Protocol)	Often allows anonymous login, sends data in plain text
22	TCP	SSH (Secure Shell)	Remote login, target for brute force
23	TCP	Telnet (Teletype Network)	Insecure, plain text remote login
25	TCP	SMTP (Simple Mail Transfer Protocol)	Email sending, used in phishing and spam
53	UDP/TCP	DNS (Domain Name System)	Name lookup, used in DNS tunneling and spoofing
80	TCP	HTTP (HyperText Transfer Protocol)	Web traffic in plain text
110 / 143	TCP	POP3 (Post Office Protocol version 3) / IMAP (Internet Message Access Protocol)	Email retrieval
139 / 445	TCP	SMB (Server Message Block)	File sharing, many famous exploits (EternalBlue)
443	TCP	HTTPS (HyperText Transfer Protocol Secure)	Encrypted web traffic
3306	TCP	MySQL (My Structured Query Language)	Database, should never be public
3389	TCP	RDP (Remote Desktop Protocol)	Windows remote desktop, common ransomware entry

TCP = Transmission Control Protocol, UDP = User Datagram Protocol.

6. TCP (Transmission Control Protocol) vs UDP (User Datagram Protocol)

	TCP (Transmission Control Protocol)	UDP (User Datagram Protocol)
Connection	Connection-oriented (handshake first)	Connectionless
Reliability	Guarantees delivery and order	No guarantee
Speed	Slower, more overhead	Faster, lightweight
Used for	Web, SSH (Secure Shell), email, file transfer	DNS (Domain Name System), video calls, streaming, gaming
Scanning note	Easy to scan reliably	Slower and less certain to scan

7. The TCP three-way handshake

Before TCP (Transmission Control Protocol) sends data, both sides agree to connect. SYN means Synchronize, ACK means Acknowledgment, and SYN-ACK means Synchronize-Acknowledgment.

```
Client                          Server
  |  ----- SYN ----------->     |   "I want to connect"
  |  <---- SYN-ACK --------     |   "OK, I'm ready"
  |  ----- ACK ----------->     |   "Great, connected"
  |  ====== data flows =====    |
  |  ----- FIN / ACK ----->     |   connection closed
(SYN = Synchronize, ACK = Acknowledgment, FIN = Finish)
```

- **SYN (Synchronize) flood attack:** an attacker sends many SYNs and never completes the handshake, so the server runs out of resources.
- **Nmap (Network Mapper) link:** a default SYN scan (nmap -sS) sends a SYN. If it receives SYN-ACK, the port is **open**. If it receives RST (Reset), the port is **closed**. No reply usually means **filtered** by a firewall.
- **Half-open scan:** after SYN-ACK, Nmap sends RST (Reset) instead of the final ACK, so the connection is never completed.

8. Other protocols you must know

- **ARP (Address Resolution Protocol):** maps an IP (Internet Protocol) address to a MAC (Media Access Control) address on the local network. It has no authentication, so ARP spoofing is possible.
- **ICMP (Internet Control Message Protocol):** used by ping and traceroute for diagnostics.
- **DHCP (Dynamic Host Configuration Protocol):** automatically gives devices an IP address, gateway and DNS (Domain Name System) server.
- **DNS (Domain Name System):** turns names into IP addresses. We cover it fully on Day 2.
- **HTTP (HyperText Transfer Protocol) / HTTPS (HyperText Transfer Protocol Secure):** the web. We cover it fully on Day 2.

9. Quick subnetting primer

A subnet mask shows which part of an IP (Internet Protocol) address is the network and which part is the host. CIDR (Classless Inter-Domain Routing) notation writes it as a slash number.

CIDR (Classless Inter-Domain Routing)	Subnet mask	Usable hosts	Example
/8	255.0.0.0	16,777,214	10.0.0.0/8
/16	255.255.0.0	65,534	172.16.0.0/16
/24	255.255.255.0	254	192.168.1.0/24
/30	255.255.255.252	2	Point-to-point links

Formula: usable hosts = $2^{(32 - \text{prefix})} - 2$. Example: /24 gives $2^8 - 2 = 254$. You will use this when you tell Nmap (Network Mapper) to scan a range like 192.168.1.0/24.

10. When to use which tool (today's tools)

Tool	Use it when	Starter command
ip / ifconfig (interface configuration)	You want your own IP (Internet Protocol) address, MAC (Media Access Control) address and interfaces	ip a
ping (Packet Internet Groper)	Checking if a host is reachable (uses ICMP, Internet Control Message Protocol)	ping 8.8.8.8
traceroute	Seeing the path (hops) to a destination	traceroute google.com
netstat (network statistics) / ss (socket statistics)	Seeing open ports and connections on your machine	ss -tuln
arp	Viewing the IP to MAC table (ARP = Address Resolution Protocol)	arp -a
Wireshark	Capturing and inspecting packets	Start a capture on your Wi-Fi (Wireless Fidelity) or Ethernet interface

11. Hands-on lab for today (about 90 minutes)

Part A: Explore your own machine (20 min)

```
ip a                # find your IP, MAC, and interface name
ip route            # find your default gateway
ss -tuln            # list listening ports on your machine
ping -c 4 8.8.8.8   # test connectivity
traceroute google.com
```

Write down: your private IP (Internet Protocol) address, your gateway, your MAC (Media Access Control) address, and how many hops it takes to reach google.com.

Part B: First Wireshark capture (40 min)

- Install Wireshark (sudo apt install wireshark on Kali or Ubuntu). Select your active interface and click Start.
- In a browser, open http://neverssl.com (plain HTTP, HyperText Transfer Protocol) and also open any HTTPS (HyperText Transfer Protocol Secure) site.
- Stop the capture. In the filter bar try: `tcp.flags.syn == 1`, then `icmp`, then `http`.
- Find a full three-way handshake (SYN, SYN-ACK, ACK) and screenshot it.
- Right-click an HTTP packet, choose Follow, then TCP (Transmission Control Protocol) Stream. Notice that plain HTTP content is readable. Compare with HTTPS, where the payload is

encrypted.

Part C: TryHackMe (30 min)

- Create a free TryHackMe account.
- Start the **Pre Security** path, then do the 'Introductory Networking' room (OSI model, packets and frames, TCP/IP).
- Rule: do not move to the next room until you can explain the previous one in your own words.

12. Quiz review: common mistakes to avoid

- **Use the proper names:** SYN (Synchronize), SYN-ACK (Synchronize-Acknowledgment), ACK (Acknowledgment), not just request, ok, connect.
- **SYN-ACK = port OPEN** (a service is listening), not 'active connection'. **RST (Reset) = closed. No reply = filtered** (a firewall is dropping packets).
- **Nmap (Network Mapper) SYN scan is half-open:** after SYN-ACK it sends RST instead of ACK, so the connection is never completed.
- **A port identifies the program** on a machine; it does not 'give access' to the device.
- **MAC (Media Access Control) addresses work only on the local network;** IP (Internet Protocol) addresses route across networks.
- **TCP (Transmission Control Protocol) reliability = handshake + sequence numbers + acknowledgements + retransmission,** not the handshake alone.
- **HTTP (HyperText Transfer Protocol) risk:** anyone on the same network can capture passwords, cookies and form data, and can modify the page in transit. HTTPS (HyperText Transfer Protocol Secure) encrypts the content.

13. Self-test (answer without looking)

- Name the 4 TCP/IP (Transmission Control Protocol / Internet Protocol) layers and one protocol in each.
- What is the difference between a MAC (Media Access Control) address and an IP (Internet Protocol) address?
- What does the port number identify?
- Draw the TCP (Transmission Control Protocol) three-way handshake from memory.
- Why is TCP port scanning more reliable than UDP (User Datagram Protocol) scanning?
- How many usable hosts are in a /24 network?
- Why is plain HTTP (HyperText Transfer Protocol) dangerous on public Wi-Fi (Wireless Fidelity)?

14. Interview-ready answers

- **TCP vs UDP?** TCP (Transmission Control Protocol) is reliable and connection-oriented with a handshake, sequence numbers and retransmission. UDP (User Datagram Protocol) is fast and connectionless with no delivery guarantee.
- **What is a port?** A logical number that identifies a specific service on a host. The IP (Internet Protocol) address finds the machine and the port finds the program.

- **What happens in a three-way handshake?** SYN (Synchronize), SYN-ACK (Synchronize-Acknowledgment), ACK (Acknowledgment), after which the connection is established.

15. Day 1 checklist

- [ ] Ran the 5 commands on my machine and noted my IP (Internet Protocol) address, gateway and MAC (Media Access Control) address
- [ ] Captured traffic in Wireshark and found a handshake
- [ ] Finished the TryHackMe Introductory Networking room
- [ ] Can answer all 7 self-test questions
- [ ] Wrote a half-page note of what I learned (for my portfolio)

Tomorrow (Day 2): DNS (Domain Name System), HTTP (HyperText Transfer Protocol), HTTPS (HyperText Transfer Protocol Secure) and TLS (Transport Layer Security), and how the web really works.

Glossary: all abbreviations and full forms

Short form	Full form
ACK	Acknowledgment
ARP	Address Resolution Protocol
CIDR	Classless Inter-Domain Routing
DHCP	Dynamic Host Configuration Protocol
DNS	Domain Name System
FIN	Finish
FTP	File Transfer Protocol
HTTP	HyperText Transfer Protocol
HTTPS	HyperText Transfer Protocol Secure
ICMP	Internet Control Message Protocol
IMAP	Internet Message Access Protocol
IP	Internet Protocol
IPv4	Internet Protocol version 4
JPEG	Joint Photographic Experts Group
JSON	JavaScript Object Notation
MAC	Media Access Control
NAT	Network Address Translation
Nmap	Network Mapper
OSI	Open Systems Interconnection
POP3	Post Office Protocol version 3

Short form	Full form
RDP	Remote Desktop Protocol
RST	Reset
SMB	Server Message Block
SMTP	Simple Mail Transfer Protocol
SQL	Structured Query Language
SSH	Secure Shell
SSL	Secure Sockets Layer
SYN	Synchronize
SYN-ACK	Synchronize-Acknowledgment
TCP	Transmission Control Protocol
TCP/IP	Transmission Control Protocol / Internet Protocol
TLS	Transport Layer Security
UDP	User Datagram Protocol
Wi-Fi	Wireless Fidelity
XSS	Cross-Site Scripting